

3.3 Topic 5: On the wild side

Students will be assessed on their ability to:

- 1 Demonstrate knowledge and understanding of the *How Science Works* areas listed in the table on page 12 of this specification.
- 2 Describe the structure of chloroplasts in relation to their role in photosynthesis.
- 3 Describe the overall reaction of photosynthesis as requiring energy from light to split apart the strong bonds in water molecules, storing the hydrogen in a fuel (glucose) by combining it with carbon dioxide and releasing oxygen into the atmosphere.
- 4 Describe the light-dependent reactions of photosynthesis including how light energy is trapped by exciting electrons in chlorophyll and the role of these electrons in generating ATP, and reducing NADP in photophosphorylation and producing oxygen through photolysis of water.
- 5 Describe how phosphorylation of ADP requires energy and how hydrolysis of ATP provides an immediate supply of energy for biological processes.
- 6 Describe the light-independent reactions as reduction of carbon dioxide using the products of the light-dependent reactions (carbon fixation in the Calvin cycle, the role of GP, GALP, RuBP and RUBISCO) and describe the products as simple sugars that are used by plants, animals and other organisms in respiration and the synthesis of new biological molecules (including polysaccharides, amino acids, lipids and nucleic acids).
- 7 Carry out calculations of net primary productivity and explain the relationship between gross primary productivity, net primary productivity and plant respiration.
- 8 Calculate the efficiency of energy transfers between trophic levels.
- 9 Discuss how understanding the carbon cycle can lead to methods to reduce atmospheric levels of carbon dioxide (including the use of biofuels and reforestation).

- 10 Explain that the numbers and distribution of organisms in a habitat are controlled by biotic and abiotic factors.
- 11 **Describe how to carry out a study on the ecology of a habitat to produce valid and reliable data (including the use of quadrats and transects to assess abundance and distribution of organisms and the measurement of abiotic factors, e.g. solar energy input, climate, topography, oxygen availability and edaphic factors).**
- 12 Explain how the concept of niche accounts for distribution and abundance of organisms in a habitat.
- 13 Describe the concept of succession to a climax community.
- 14 Outline the causes of global warming – including the role of greenhouse gases (carbon dioxide and methane, CH₄) in the greenhouse effect.
- 15 Describe the effects of global warming (rising temperature, changing rainfall patterns and seasonal cycles) on plants and animals (distribution of species, development and life cycles).
- 16 Explain the effect of increasing temperature on the rate of enzyme activity in plants, animals and micro-organisms.
- 17 **Describe how to investigate the effects of temperature on the development of organisms (e.g. seedling growth rate, brine shrimp hatch rates).**
- 18 Analyse and interpret different types of evidence for global warming and its causes (including records of carbon dioxide levels, temperature records, pollen in peat bogs and dendrochronology) recognising correlations and causal relationships.
- 19 Describe that data can be extrapolated to make predictions, that these are used in models of future global warming, and that these models have limitations.

- 20 Discuss the way in which scientific conclusions about controversial issues, such as what actions should be taken to reduce global warming or the degree to which humans are affecting global warming, can sometimes depend on who is reaching the conclusions.
- 21 Describe how evolution (a change in the allele frequency) can come about through gene mutation and natural selection.
- 22 Explain how reproductive isolation can lead to speciation.
- 23 Describe the role of the scientific community in validating new evidence (including molecular biology, e.g. DNA, proteomics) supporting the accepted scientific theory of evolution (scientific journals, the peer review process, scientific conferences).